

Customer Data Analysis: Summary Report

2,100 customers, one purchase each, from 27 Oct 2022 to 23 Jul 2025 · source: Customers_Fakedata.csv (2,150 raw rows, 50 duplicates removed) · tools: Python (pandas, SciPy, matplotlib), an interactive HTML dashboard and Power BI

Customers 2,100 unique IDs	Recorded revenue \$1.02M 2,003 priced orders	Average order \$510 median \$519	Purchases / month 59.9 \$29.3K / month
Average rating 3.03 / 5 1,487 valid ratings	Satisfied (4-5 stars) 39% of valid ratings	Dissatisfied (1-2 stars) 39% of valid ratings	Complete records 9.7% valid in all 6 fields

Summary

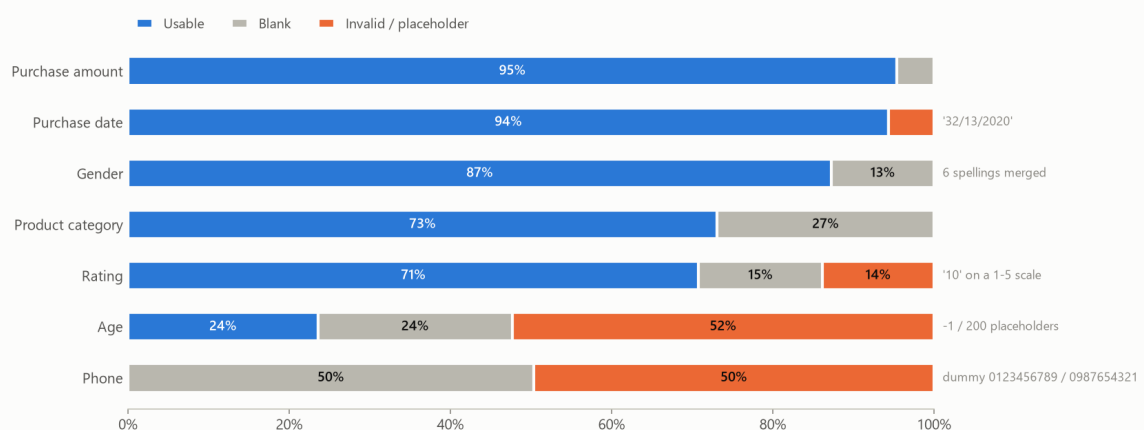
- **The biggest finding is the data itself.** Only 9.7% of records are complete: age is usable for 24% of customers, 27% of purchases have no category, and no phone number is real.
- **Big tickets carry the revenue.** \$1.02M in recorded revenue; \$750+ orders are 26% of orders but 45% of revenue.
- **Satisfaction is split down the middle.** Average 3.03 / 5: 39% dissatisfied, 39% satisfied, and the same in every category, gender, age group and order size.
- **Sales are flat.** About 60 purchases and \$29K a month for 32 months (trend $p = 0.48$); January-June revenue +1.3% in 2024 and -2.6% in 2025. The apparent August-September dip is a calendar artefact.
- **No segment behaves differently.** None of 14 tests on gender, age, category, weekday or email provider is significant. The file behaves like randomly generated data, so this is the expected result.

How the data was cleaned

50 exact duplicate rows and 2 redundant columns were removed. Impossible values were blanked, not guessed: age placeholders -1 and 200 (1,099 rows), rating '10' on a 1-5 scale (291), the date '32/13/2020' (118). Six spellings of gender were merged, and blank categories became 'Unknown'. Missing amounts and ages were **not imputed**, because imputing would invent 5% of revenue and 76% of ages. Each metric uses every row that is valid for it. This is safe because the gaps are independent of each other (maximum correlation 0.04). Details: DATA_QUALITY_REPORT.md.

Only 9.7% of customer records are complete

Share of the 2,100 customers with a usable value in each field, after removing 50 duplicate rows



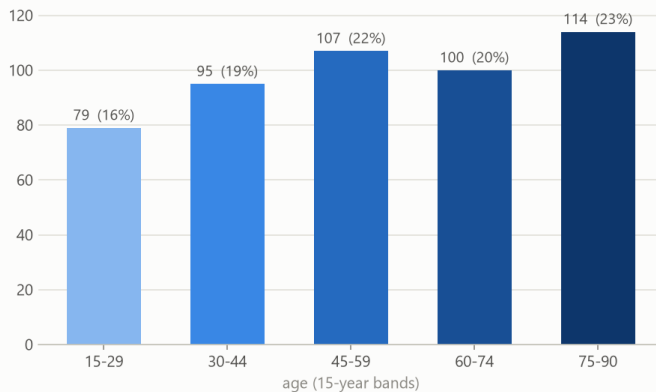
Finding 1: the biggest finding is the data itself

Age is the weakest field: only 495 of 2,100 customers have a real age. The rest are the placeholders -1 or 200, or blank. Gender is recorded for 87%, but it contradicts the first name in 50% of rows, so neither field can yet support targeting.

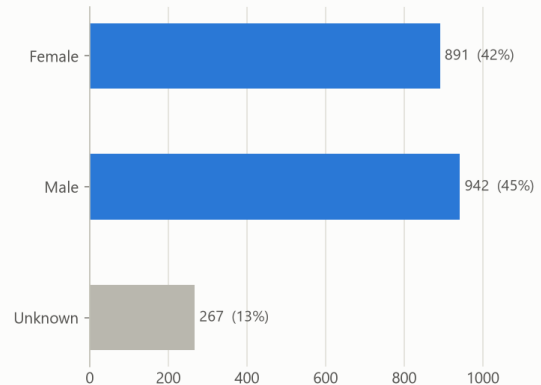
Customers span every age from 15 to 90 (median 55); genders are balanced

Age is usable for only 24% of customers (the rest are -1, 200 or blank). Gender is 49% female / 51% male where recorded

Customers by age group (the 495 with a real age)



Customers by gender



- Real ages cover every value from 15 to 90 (median 55), spread evenly across five 15-year bands.
- Recommendation: collect a date of birth, not a free-text age, and use drop-down lists for gender, category and rating.

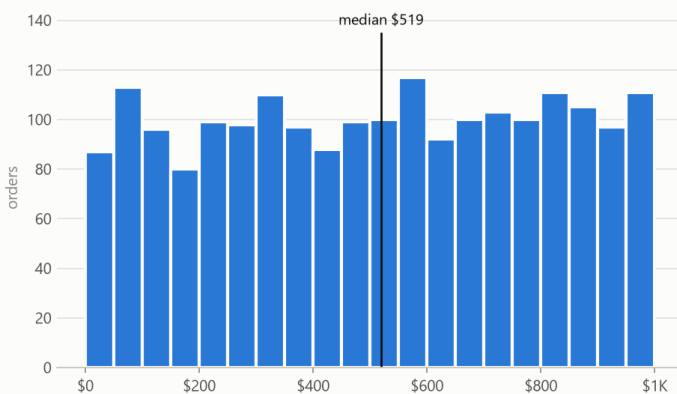
Finding 2: big tickets carry the revenue

Order values are spread evenly from \$5 to \$1,000, so there is no typical basket. Because of that, the biggest orders dominate revenue: one order in four is \$750+, and those orders bring 45% of revenue.

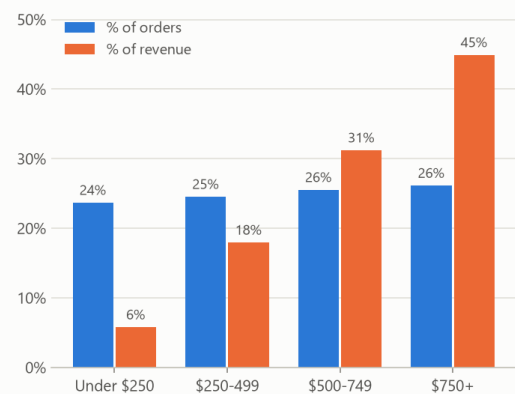
\$750+ orders are 26% of orders but 45% of revenue

Spending is spread evenly from \$5 to \$1000 (average \$510); total recorded revenue \$1.02M. 97 orders have no amount

Purchase amounts (2,003 priced orders, \$50 bins)



Orders vs revenue by order size



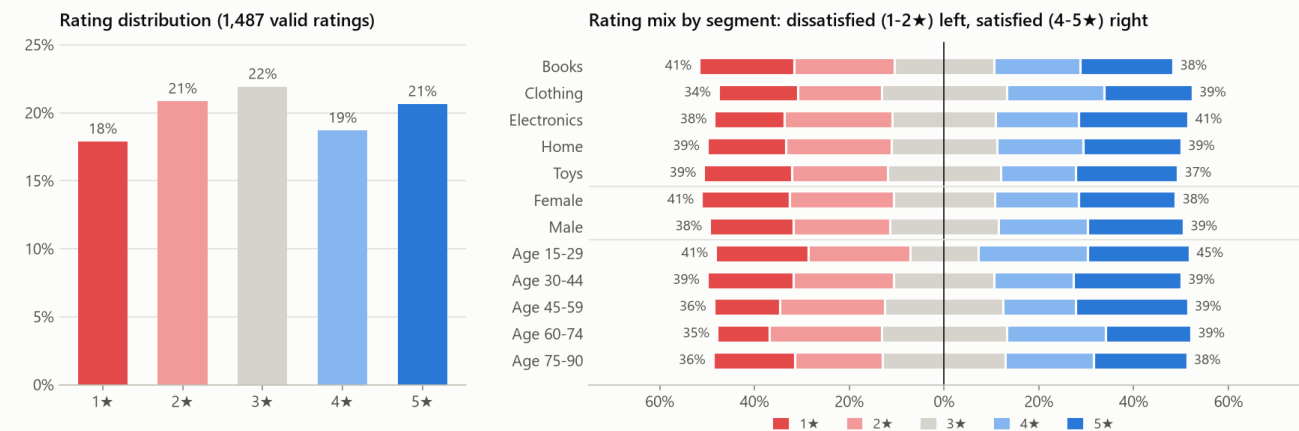
- An average \$750+ order (\$876) is worth seven average orders under \$250 (\$125).
- Recommendation: give big orders priority service, and A/B-test bundles or upgrades on \$250-749 orders.

Finding 3: as many unhappy customers as happy ones

All five star levels get about a fifth of the ratings. The split is the same in every category, gender, age group and order size, and the rating does not depend on how much was spent ($\rho = -0.027$).

As many customers are unhappy (39%) as happy (39%): average 3.03 / 5

Every star level gets about a fifth of ratings, and the mix is the same in every category, gender and age group (all $p > 0.11$)



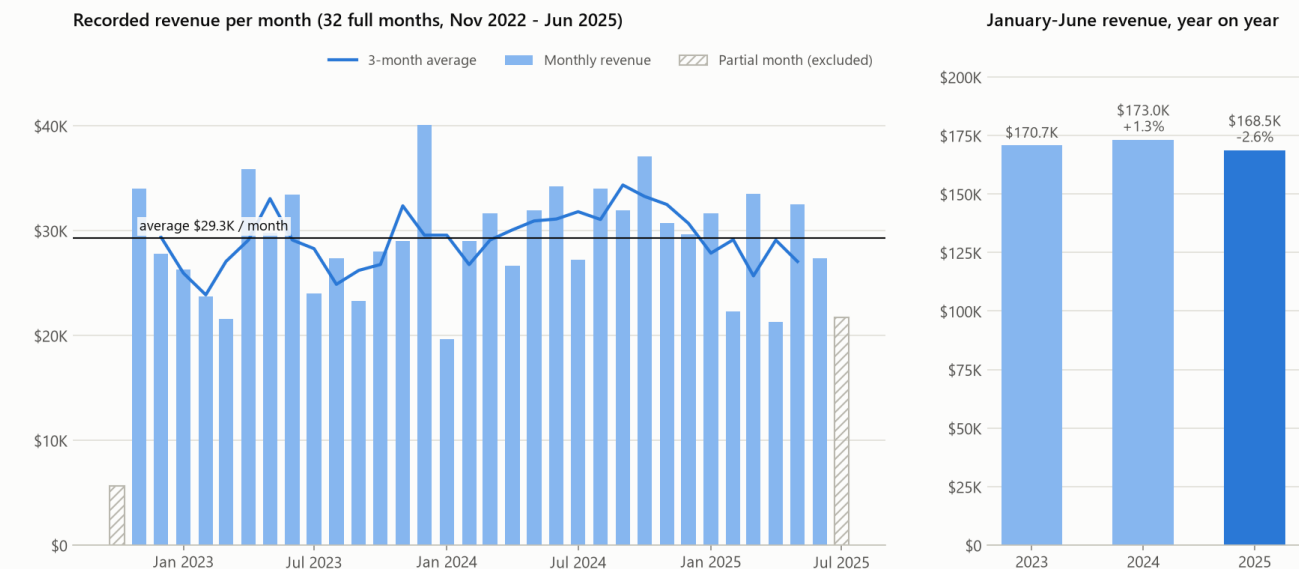
- Dissatisfaction is not explained by any product, price or customer group recorded here. The cause is outside the data, for example delivery, service or product quality.
- Recommendation: add a reason code to every rating, and follow up every 1-2 star rating within 48 hours (target: dissatisfied share below 25%, average rating 3.5).

Finding 4: sales are flat

Revenue swings between about \$20K and \$40K a month around a flat average of \$29.3K, with no upward or downward trend over 32 complete months.

Sales are flat: about 60 purchases and \$29K a month, with no trend

Trend slope +0.12 purchases per month ($p = 0.48$); the same half-year moved +1.3% then -2.6%. The Oct 2022 and Jul 2025 bars are partial months



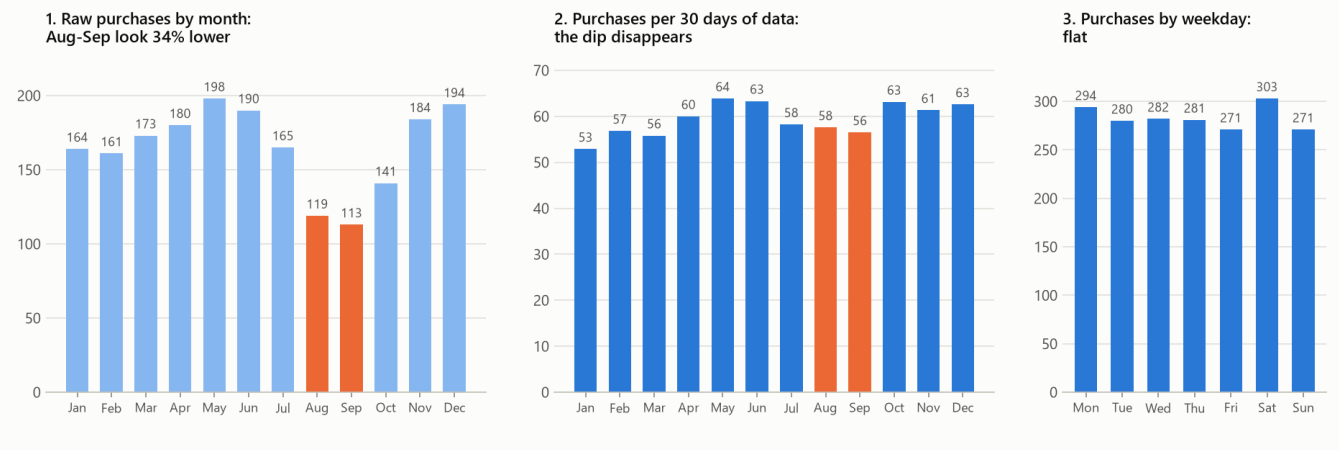
- Like-for-like January-June revenue: \$170.7K (2023), \$173.0K (2024) and \$168.5K (2025). The business is stable, but growth will need a deliberate action.

Finding 5: no seasonality, but two traps that look like patterns

In raw totals, August and September look about a third weaker than other months, and 2025-Q3 looks like a collapse. Both are artefacts: August and September occur in only 2 of the years covered, and 2025-Q3 holds only 23 days of data. Measured per 30 days of data, every month is alike.

No seasonal or weekly pattern once calendar coverage is accounted for

The data runs 2022-10-27 to 2025-07-23: Aug and Sep occur in 2 of the years covered, Jan-Jun in 3. Adjusted, months differ only by chance ($p = 0.82$); so do weekdays ($p = 0.82$)



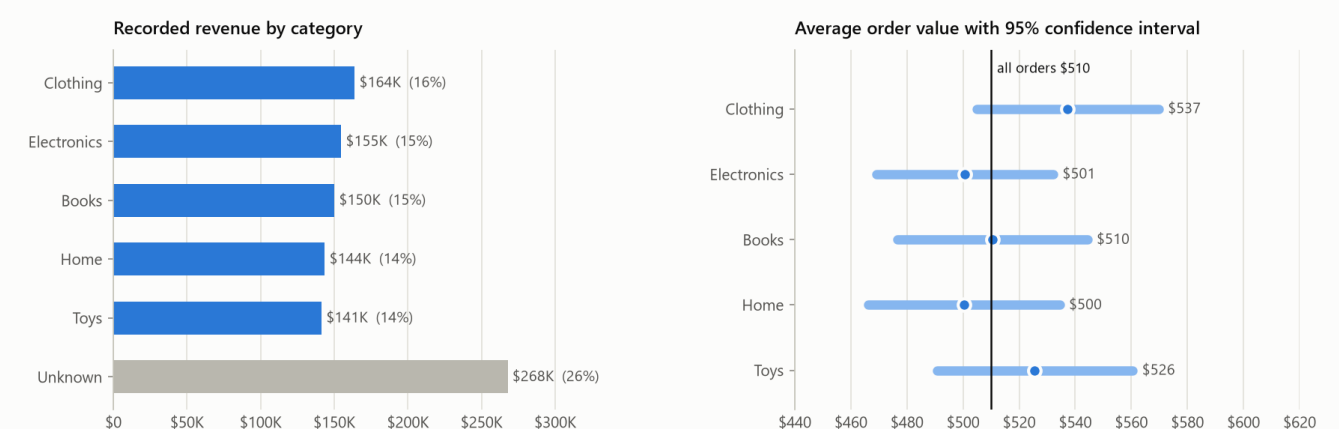
- Weekdays are also flat (271-303 purchases each). Reports on this data should always use complete periods and compare rates, not totals.

Finding 6: no category or customer segment stands out

Each of the five categories brings 14-16% of revenue. Clothing leads, but order values do not differ between categories ($p = 0.45$). 26% of revenue (\$268K) has no category at all.

No category leads: each has 14-16% of revenue, and 26% is unassigned

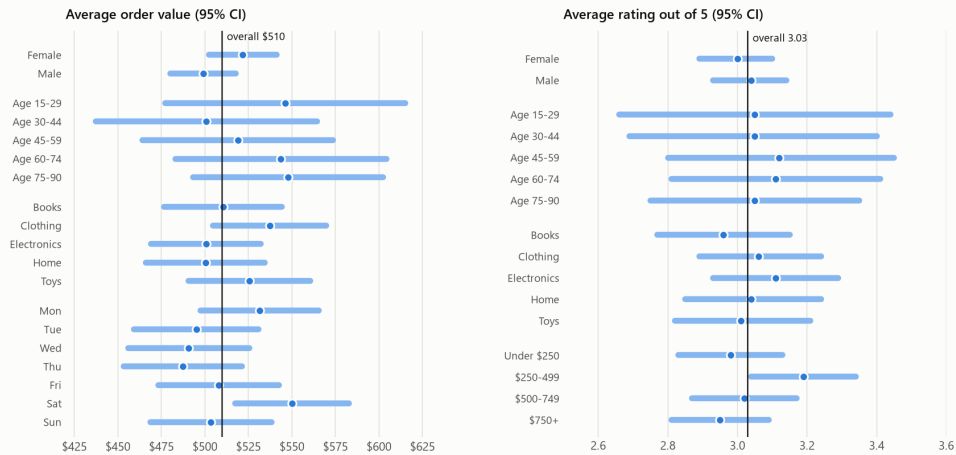
Clothing is first by a margin chance can explain (order value $p = 0.45$). 565 purchases (\$268K) have no category



- The same holds for gender, age, weekday and email provider (see the next chart): every difference is within chance.

No customer segment spends or rates differently from the rest

None of the 14 segment tests is significant (smallest $p = 0.09$). 2 of 35 intervals miss the overall line (Sat, \$250-499): chance alone predicts about 1.8



Each dot is a segment average and each bar its 95% confidence interval. 2 of 35 intervals miss the overall line, about what chance alone produces (1.8).

Recommendations

#	Action	Measure it with
1	Fix data capture at the source: date of birth instead of free-text age; drop-downs for gender, category and rating (1-5 only); a date picker; phone validation; no duplicate IDs on import.	Complete records 9.7% → 90%+
2	Back-fill a category for the 565 uncategorised purchases (\$268K of revenue) from the order or product system.	Revenue with no category 26% → 0%
3	Find out why 2 in 5 customers are dissatisfied: add a reason code and comment to each rating, link ratings to delivery and returns data, and follow up every 1-2 star rating within 48 hours.	Avg rating 3.03 → 3.5; dissatisfied 39% → <25%
4	Protect and grow big orders: priority service for \$750+ orders (45% of revenue), and A/B-tested bundles or upgrades on \$250-749 orders. Use experiments, not demographic targeting, because no segment behaves differently.	Share of \$750+ orders; AOV
5	Start measuring loyalty: one persistent customer ID across orders, so that repeat rate and lifetime value become measurable. Track monthly purchases against the ~60 / month baseline.	Repeat-purchase rate; purchases / month

Limitations

- Each customer has exactly one purchase, so there is no repeat-purchase, retention or lifetime-value analysis.
- Age results rest on 495 customers (24%), and gender contradicts the first name in half of the rows.
- Revenue is recorded revenue: 97 purchases (about \$49K) have no amount, and they were not estimated.
- The file behaves like randomly generated data: amounts uniform from \$5 to \$1,000 (KS $p = 0.50$), ages uniform from 15 to 90 ($p = 0.16$), ratings equally likely ($p = 0.09$), gender unrelated to the first name ($p = 0.12$), and exactly a 1,000-day window. "No difference" therefore describes this file, not a real customer base.

Deliverables

- Customer_Dashboard.html:** an interactive dashboard (8 KPIs, 14 charts, a segment scorecard, 9 filters plus click-to-filter). It opens offline in any browser.
- Customer_Dashboard.pbix:** the same analysis as a 5-page Power BI report with 30 DAX measures. Also included: KEY_INSIGHTS.md, DATA_QUALITY_REPORT.md, Cleaned_Customers.csv, charts/ and the Python scripts that rebuild everything from the raw file.